ECONOMIC GROWTH
SPRING 2020
EXERCISE 1

1. Please, explain what the two (equally valid) conditions are for endogenous growth to arise in the one sector setting.

2. In a one sector setting, is endogenous growth feasible if we invoke the following production technologies? *Explain.*

$$Y = F(K, L) \equiv AK, \ A > 0$$

$$Y = F(K, L) \equiv AK^{\alpha}L^{1-\alpha} + BK, \ A > 0 \ \wedge \ B > 0. \tag{1}$$

$$Y = F(K, L) \equiv \left(\delta L^{\frac{E-1}{E}} + (1-\delta) K^{\frac{E-1}{E}}\right)^{\frac{E}{E-1}}, \ E > 0.$$

3. Consider a Diamond model, where people live for two periods – work during first period ("youth") , and consume their total savings during second period ("old age"). Assume the aggregate production function fulfills

$$\lim_{k \to \infty} f'(k) = A > 0,$$

where $f'(k)$ is the marginal product of capital (k). Aside from the above feature, the production function is neoclassical, and involves only capital and labor input. There is no exogenous technological change. Will perpetual growth be possible in this economy? Why/Why not?

4. Suppose a researcher is interested in examining the impact of the investment rate, $\ln(s_{Kt})$, on growth in output per capita, $\ln(y_{t+1}) - \ln(y_t)$. As a result he/she runs the regression (using a panel of countries, where i denotes "country")

$$\ln(y_{it+1}) - \ln(y_{it}) = a + b \ln y_{it} + c \ln(s_{iKt}) + \epsilon_i,$$

and ϵ is a noise term. The result is that c (the main parameter of interest) is insignificant. The researcher concludes that investment does not affect the growth rate of the economy. Do you (necessarily) agree with this conclusion? Why/Why not?

Consider an economy in the process of development. It is a one good economy, where output can be either consumed or invested. The economy is closed, and the size of the population is constant at L. The is no capital depreciation. All markets are competitive. Time is continous.

The representative household maximizes utility from consumption over an infinite horizon $\int_0^\infty \ln c_t e^{-\rho t} dt$, subject to a non-negativity constraints on consumption; its wealth accumulation rule

$$\dot{a}_t = (1-\tau) r a_t + (1-\tau) w_t - c_t, a_0 \text{ given},$$

and, the standard No-Ponzi game condition. a is wealth, τ is an income tax (assumed constant through time), r is the real rate of interest, w the wage (labor supply is exogenous; the household supplies a unit of labor inelastically) and c is consumption. A nessesary condition for an optimal solution is that $\dot{c}/c = (1-\tau) r - \rho$. The representative firm is equipped with the following production technology

$$Y_i = f(G) K^\alpha L^{1-\alpha},$$

where $f(.)$ will be specified below, and is taken as given by the representative firm.

Q1. Solve the representative firm's profit maximization problem

Q2. Assume the government balances its budget. Show that this implies:

$$G = \tau Y$$

From now on, assume:

$$f(G) = \left(\frac{G}{K^\phi L^\delta} \right)^\gamma, \tag{2}$$

Q3. Comment on equation (2): What could be an interpretation of the presence of $K^\phi L^\delta$ in the denominator?

Q4. Under what parameter restriction will the model exhibit endogenous growth?

Q5. Assuming the restriction required for endogenous growth is fulfilled, derive an expression for the growth rate of GDP (per capita) as a function of the tax rate. How does increasing taxes impact on the growth rate?

Q6. Will it be optimal, in this model, to finance G by way of lump-sum taxes rather than via an income tax? Explain why/why not.

Q7. Endogenous growth models often predict that larger economies will grow faster than smaller economies, a phenomenon labelled "scale effects". Does the present model exhibit scale effects? What does the evidence say?